Higher National Diploma in Engineering (Electrical/Mechanical/Civil/BSE) First Year, First Semester Examination - 2016

MA 1101: Engineering Mathematics

Time: Two hours

Question - 01

i) Prove that

$$\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$$

(6 Marks)

ii) If matrix

$$A = \frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$$

prove that A−1=AT, AT being the transpose of A.

(7 Marks)

iii) Find the Eigen values and the corresponding Eigen vectors of the following matrix

$$A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$$

(12 Marks)

(Total 25 Marks)

Question - 02

i) Test the following series for convergence.

$$\sum_{1}^{\infty} \frac{\sqrt[3]{3n^2+1}}{\sqrt[4]{4n^3+2n+7}}$$ (10 Marks)

ii) Show that following series is convergent if x<1/3 and divergent if x≥1/3

$$\sum \frac{4 \,.\, 7 \,.\ldots\ldots .(3n+1)}{1 \,.\, 2. \ldots\ldots n} x^n$$

(7Marks)

iii) If u=xyz, $v=x^2 + y^2 + z^2$, w=x + y + z then find

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)}$$

(8 Marks)

(Total 25 Marks)

Question - 03

i)

a) Evaluate

$$\lim_{\substack{x\to\infty \\ y\to 2}} \left(\frac{xy+2}{x^2+2y^2}\right)$$

b) Show that the function

$$F(x, y) = 2x^2 + y, \ (x, y) \neq (1, 2)$$
$$= 0 \qquad , (x, y) = (1, 2) \text{ is discontinuous at } (1, 2)$$

(2 x 3 = 6 Marks)

ii) If $u=e^{xyz}$, find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$

(7 Marks)

iii) Verify Euler's theorem on homogeneous function when

$f(x, y, z)=3x^2yz + 5xy^2z + 4z^4$

(5 Marks)

iv) If $u = \sin^{-1}\frac{(x^2+y^2)}{x+y}$ show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$

(7 Marks)

(Total 25 Marks)

Question - 04

i) Evaluate Ff_A xy dx dy where A is given by $x^2 + y^2-2x = 0$,$y^2=2x$ and

(12 Marks)

ii) Calculate the volume of the solid bounded by the following surfaces

z=o , and x + y + z = 3 =0. 28

(13 Marks)

(Total 25 Marks)